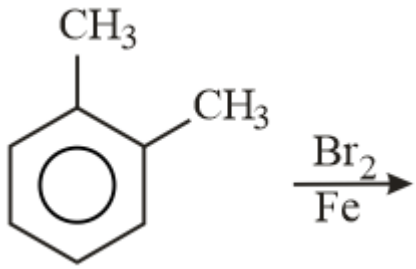
Quizrr Chapter-Wise Test for JEE Advanced - 2025

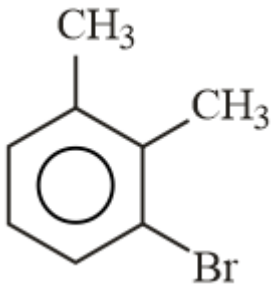
By: C I P H Ξ R

Q.1

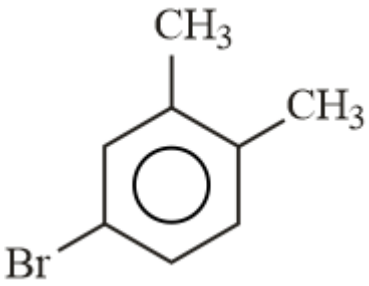


Major product is :-

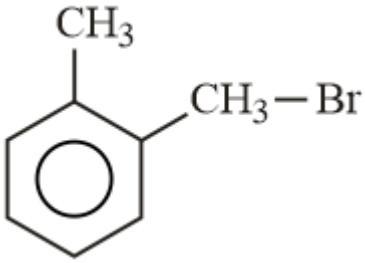
(A)



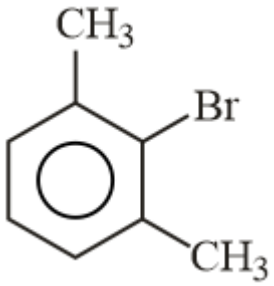
(B)



(C)



(D)



Q.2

The reaction of $SOCl_2$ on alkanols to form alkyl chlorides gives good yields because -

(A) Alkyl chlorides are immiscible with $SOCl_2$

(B) The other products of the reaction are gaseous and escape out

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(C) Alcohol and $SOCl_2$ are soluble in water

(D) The reaction occurs via intermediate formation of an alkyl chloro sulphite

Q.3

A carbon compound A forms B with sodium metal and again A forms C with PCl_5 but B and C form diethylether. Therefore A, B & C are -

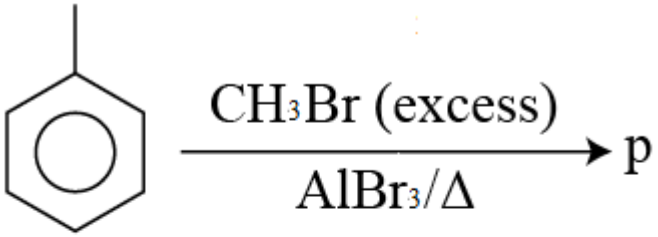
(A) $C_2H_5OH, C_2H_5ONa, C_2H_5Cl$

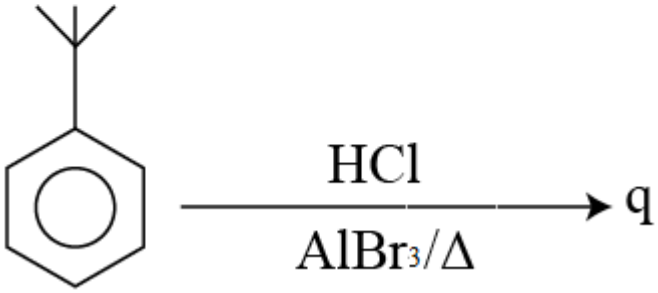
(B) $C_2H_5Cl, C_2H_5ONa, C_2H_5OH$

(C) $C_2H_5OH, C_2H_6, C_2H_5Cl$

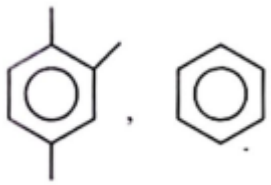
(D) $C_2H_5OH, C_2H_5Cl, C_2H_5ONa$

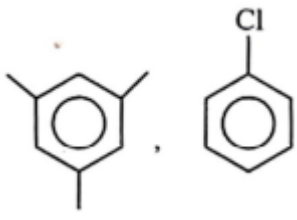
Q.4

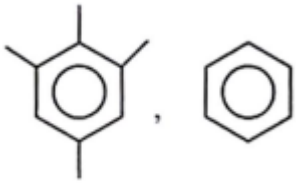




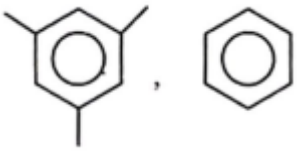
p and q from the above reactions are

(A) 

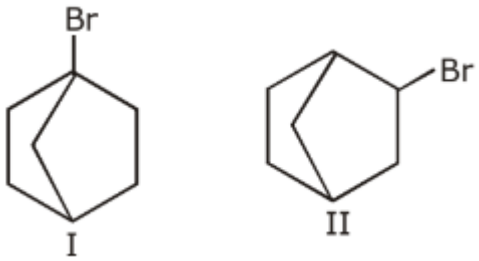
(B) 

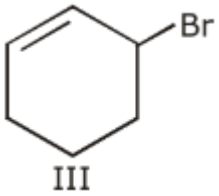
(C) 

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(D) 

Q.5





S_N2 reactivity of these substrate, under identical conditions, will be in the order as

(A) I > II > III

(B) III > II > I

(C) III > I > II

(D) II > III > I

Q.6

The yield of alkyl bromide obtained as a result of heating the dry silver salt of carboxylic acid with bromine what will be the order of formation w.r.t. alkyl bromide -

(A) $1^\circ > 3^\circ > 2^\circ$ bromides

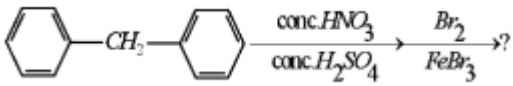
(B) $1^\circ > 2^\circ > 3^\circ$ bromides

(C) $3^\circ > 2^\circ > 1^\circ$ bromides

(D) $3^\circ > 1^\circ > 2^\circ$ bromides

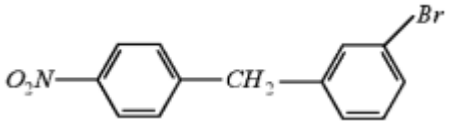
Q.7

Predict major product of the following reaction

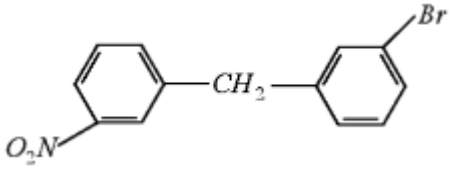


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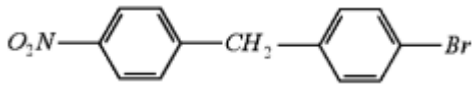
(A)



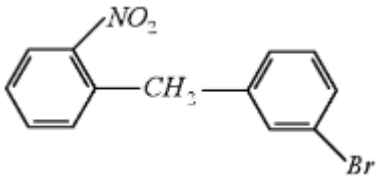
(B)



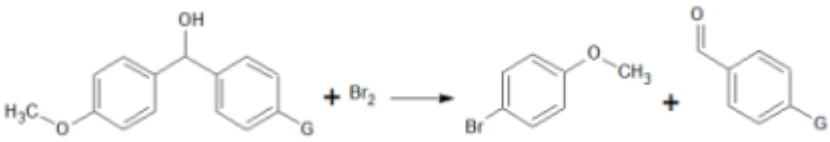
(C)



(D)



Q.8



The rate of reaction is moderately affected by the nature of G. Choose the correct decreasing rate of reaction with various substitutents as G

(A) $-NO_2 > -Br > -H > -CH_3 - OCH_3$

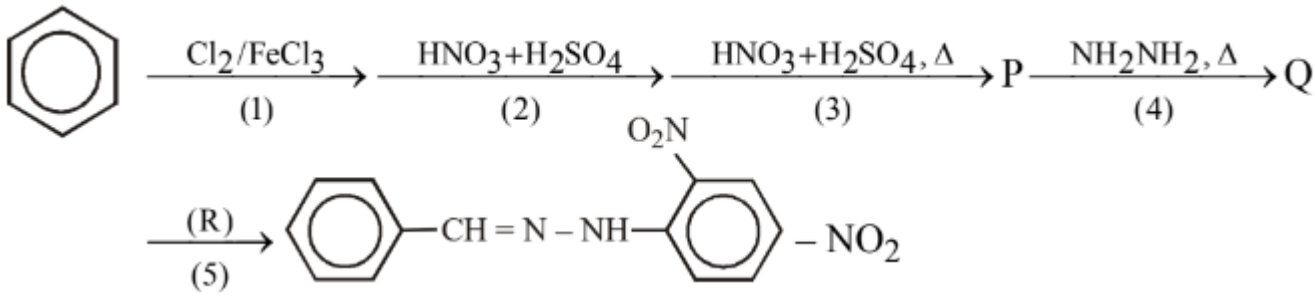
(B) $-OCH_3 > -CH_3 > -H > -Br > -NO_2$

(C) $-OCH_3 > -Br > -CH_3 > -H > -NO_2$

(D) $-NO_2 > -OCH_3 > -Br > -H > -CH_3$

Q.9

Choose the correct statements for the reactions



(A) R gives an aldol condensation reaction on heating with NaOH solution

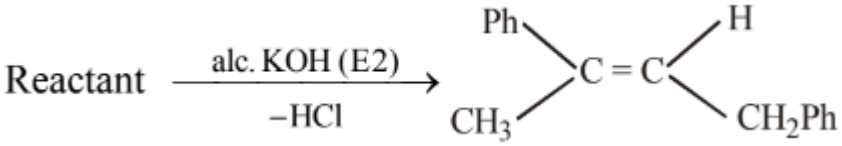
(B) The compound Q gives a yellow precipitate in acetone

(C) Step 4 is an aromatic nucleophilic substitution reaction

(D) The end product is a mixture of three compounds

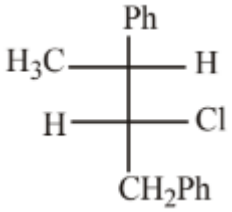
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Q.10

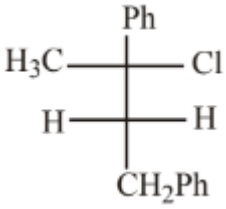


Which of the following molecule (s) is/are suitable reactant?

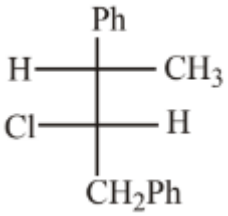
(A)



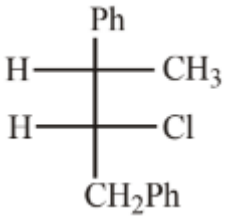
(B)



(C)



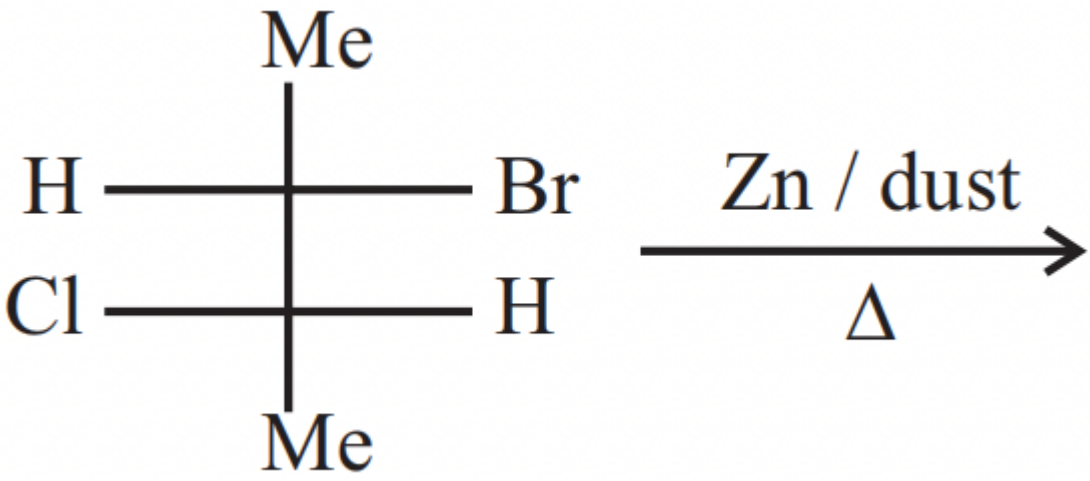
(D)



Q.11

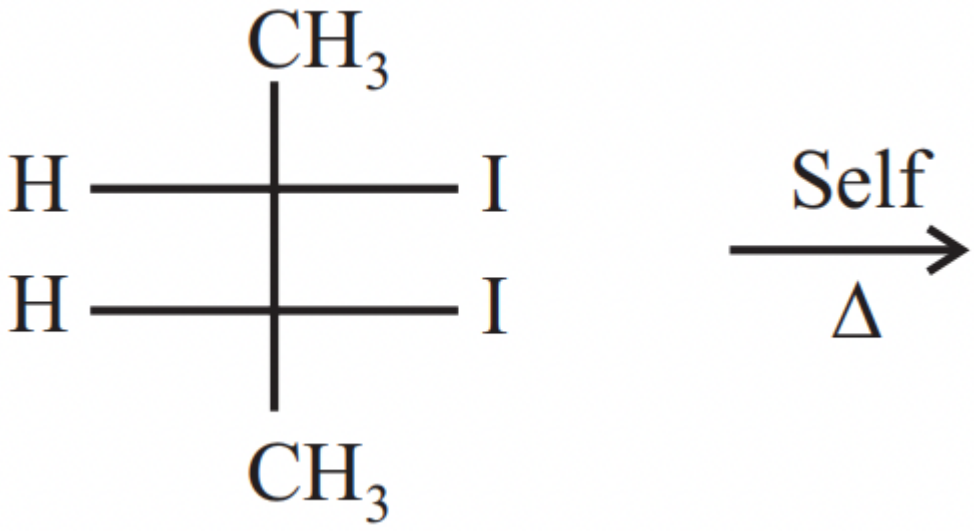
Which will be anti elimination?

(A)

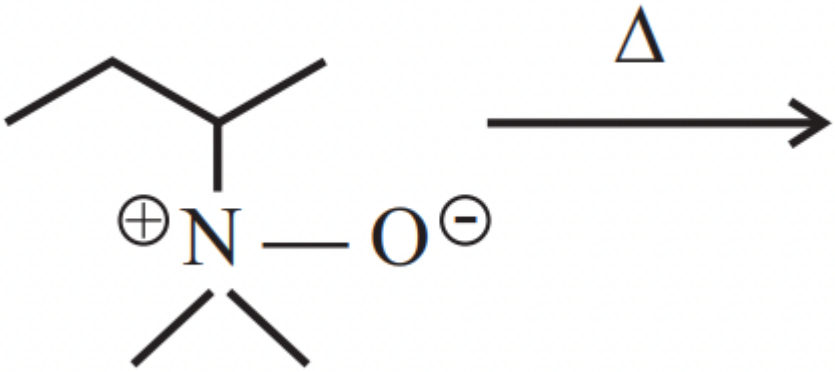


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(B)



(C)



(D) All

Q.12

Passage:

A free radical substitution reaction : $R-X \longrightarrow R-Y$ involves various steps. There must be a cleavage of the substrate RX so that R^* radicals are produced **Step I :** Spontaneous cleavage : $R-X \longrightarrow R^*+X^*$ **Step I' :** Cleavage can be caused by light or heat or more often, there is no cleavage R^* is produced by abstraction $\mathrm{R}-\mathrm{X}+\mathrm{W}^* \longrightarrow \mathrm{R}^*+\mathrm{W}-\mathrm{X}$ (W^* is produced by adding a compound, such as a peroxide, that spontaneously forms free radicals. Such as compound is called an initiator).

Once $R*$ is formed, it can go to product in two ways, by abstraction. **Step II :** $\mathrm{R}^*+\mathrm{Y}-\mathrm{W} \longrightarrow \mathrm{R}-\mathrm{Y}+\mathrm{W}^*$ or by coupling **Step III :** $\mathrm{R}^*+\mathrm{Y}^* \longrightarrow \mathrm{R}-\mathrm{Y}$

In a reaction with moderately long chain, much more of the product will be produced depending upon reactivity and type of hydrogen by abstraction. When the reaction step $\mathrm{R}-\mathrm{X} \longrightarrow \mathrm{R}^*$ takes place of a chiral carbon, racemization is almost always observed.

Question:

$$\mathrm{CH_3-C\equiv C-CH_2-CH_3} \xrightarrow{\mathrm{Br_2(1\,eq),CCl_4}} \mathrm{X} \xrightarrow{\mathrm{Br_2,h\nu}} \mathrm{Y+Z}$$

There are two isomeric forms Y and Z for the product, Y and Z are $-$

(A) constitutional isomers

(B) enantiomers

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(C) 3 enantiomers

(D) diastereomers

Q.13

Passage:

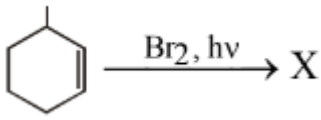
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Once $R*$ is formed, it can go to product in two ways, by abstraction. **Step II :** $\mathrm{R^* + Y - W \longrightarrow R - Y + W^*}$ or by coupling **Step III :** $\mathrm{R^* + Y^* \longrightarrow R - Y}$

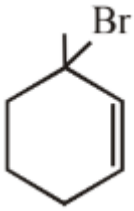
In a reaction with moderately long chain, much more of the product will be produced depending upon reactivity and type of hydrogen by abstraction. When the reaction step $\mathrm{R - X \longrightarrow R^*}$ takes place of a chiral carbon, racemization is almost always observed.

Question:

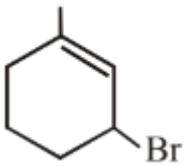
The thermodynamic product X is $-$



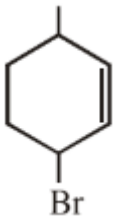
(A)



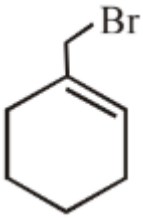
(B)



(C)



(D)



Q.14

Passage:

A free radical substitution reaction : $R - X \longrightarrow R - Y$

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involves various steps. There must be a cleavage of the substrate RX so that R^* radicals are produced **Step I :** Spontaneous cleavage : $R - X \longrightarrow R^* + X^*$ **Step I' :** Cleavage can be caused by light or heat or more often, there is no cleavage R^* is produced by abstraction $\mathrm{R - X + W^* \longrightarrow R^* + W - X}$
(W * is produced by adding a compound, such as a peroxide, that spontaneously forms free radicals. Such as compound is called an initiator).
Once $R*$ is formed, it can go to product in two ways, by abstraction. **Step II :** $\mathrm{R^* + Y - W \longrightarrow R - Y +}$ $\mathrm{W^*}$ or by coupling **Step III :** $\mathrm{R^* + Y^* \longrightarrow R - Y}$
In a reaction with moderately long chain, much more of the product will be produced depending upon reactivity and type of hydrogen by abstraction. When the reaction step $\mathrm{R - X \longrightarrow R^*}$ takes place of a chiral carbon, racemization is almost always observed.

Question:

Radical substitution is sometimes inhibited by the addition of substances like benzoquinone because

(A) benzoquinone destabilises the radical

(B) benzoquinone stabilises the alkane to be substituted

(C) benzoquinone combines with radical and produces highly stable non-productive radical

(D) benzoquinone neutralises the radical

Q.15

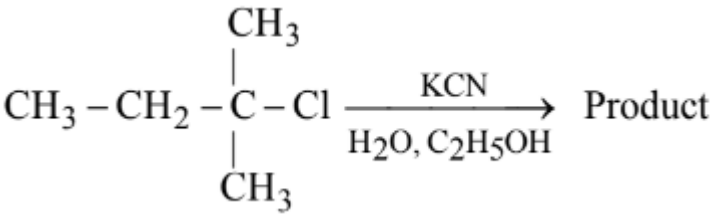
Passage:

Which atom of an amibidient nucleophile will attack a given substrate under a given set of conditions is governed by certain principles :
The principle of hard and salt acids and bases suggests that hard acids prefer hard bases and soft acids prefer soft bases. Thus, we may make a general idea that as the character of a reaction changes from S_N1 like to S_N2 like an ambident nucleophile becomes more likely to attack with its less electronegative atom.
If the negatively charged nucleophile has $\mathrm{Ag^+}$ like counter ion, which helps in removing the leaving group, characterise the reaction S_N1-like

$\mathrm{R - Cl + NaCN \rightarrow R - CN}$ less electronegative atom attack
$\mathrm{R - C + AgCN \rightarrow R - NC}$ more electronegative atom attack

The freer the nucleophile, the more likely is to attack with its more electronegative atom, but more this atom is encumbered by either solvent molecule or positive counter ions, the more likely is attack by the less electronegative atom.

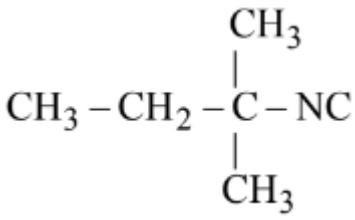
Question:



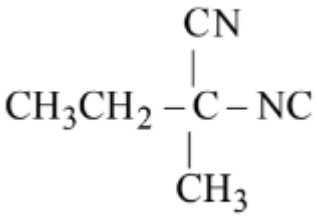
The main product is -

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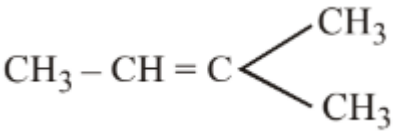
(A)



(B)



(C)



(D) None of the above

Q.16

Passage:

Which atom of an amibidient nucleophile will attack a given substrate under a given set of conditions is governed by certain principles :

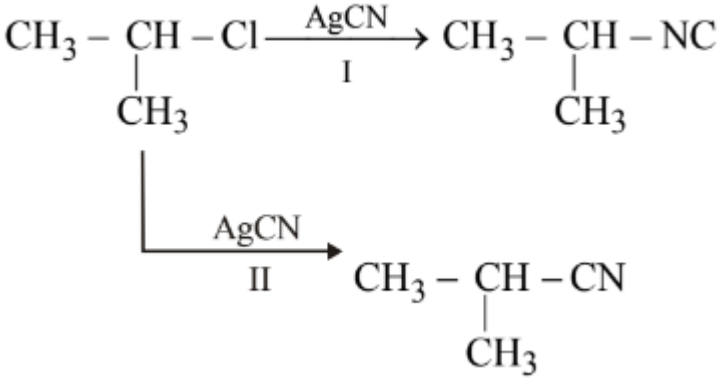
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If the negatively charged nucleophile has Ag^+ like counter ion, which helps in removing the leaving group, characterise the reaction S_N1-like

$$R-Cl+NaCN \rightarrow R-CN \quad \text{less electronegative atom attack}$$
$$R-C+AgCN \rightarrow R-NC \quad \text{more electronegative atom attack}$$

The freer the nucleophile, the more likely is to attack with its more electronegative atom, but more this atom is encumbered by either solvent molecule or positive counter ions, the more likely is attack by the less electronegative atom.

Question:



Most appropriate solvents I and II to get the maximum yield of respective products are, respectively

(A) I. H_2O, C_2H_5OH; II. Dimethyl formamide

(B) I. Dimethyl sulphoxide; II. H_2O, C_2H_5OH

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(C) I. Dimethyl formamide; II. C_2H_5OH

(D) I. H_2O, C_2H_5OH; II. Acetone

Q.17

Passage:

Which atom of an amibidient nucleophile will attack a given substrate under a given set of conditions is governed by certain principles :

The principle of hard and salt acids and bases suggests that hard acids prefer hard bases and soft acids prefer soft bases. Thus, we may make a general idea that as the character of a reaction changes from S_N1 like to S_N2 like an ambident nucleophile becomes more likely to attack with its less electronegative atom.

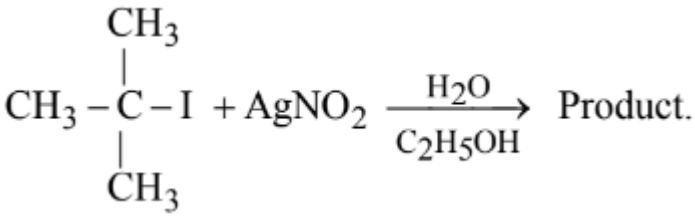
If the negatively charged nucleophile has Ag^+ like counter ion, which helps in removing the leaving group, characterise the reaction S_N1-like

$R-Cl + NaCN \rightarrow R-CN$ less electronegative atom attack

$R-C + AgCN \rightarrow R-NC$ more electronegative atom attack

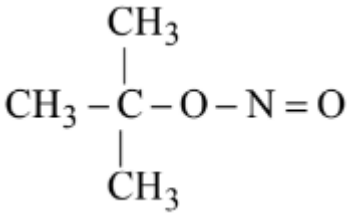
The freer the nucleophile, the more likely is to attack with its more electronegative atom, but more this atom is encumbered by either solvent molecule or positive counter ions, the more likely is attack by the less electronegative atom.

Question:

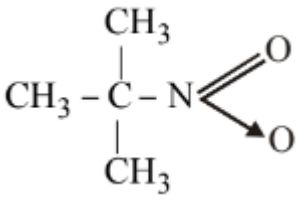


The predominant product is -

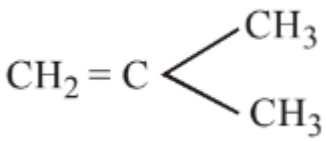
(A)



(B)



(C)

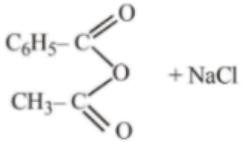
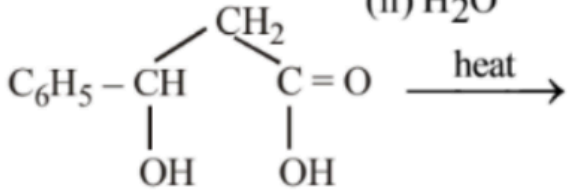


(D) None of the above

Q.18

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Match the following -

Column I (Reaction)	Column II (Product)
(A) $C_6H_5-CO-Cl \xrightarrow[(ii)\ H_2O]{(i)\ CH_2N_2}$	(p) 
(B) $C_6H_5-COOH \xrightarrow[(ii)\ H_2O]{(i)\ 2CH_3Li}$	(q) $C_6H_5-CH=CH-COOH$
(C) 	(r) $C_6H_5-CH_2-COOH$
(D) $C_6H_5-CO-Cl \xrightarrow{CH_3COONa}$	(s) $C_6H_5-CO-CH_3$
	(t) Arndt-Eistert reaction

(A) (A) - r,s, (B) - s, (C) - s, (D) - t

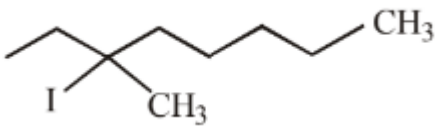
(B) (A) - r,t, (B) - s, (C) - q, (D) - p

(C) (A) - r,t, (B) - r, (C) - q, (D) - r

(D) (A) - r,t, (B) - t, (C) - r, (D) - p

Q.19

How many distinct alkene products are possible when the alkyl iodide below undergoes E2 elimination?

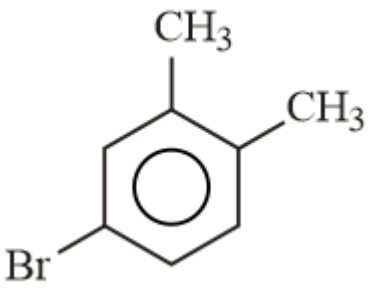


Q.20

Optically pure $(S)-(+)-2$-bromooctane $[\alpha]_D^{25^\circ}=+36^\circ$, reacts with aqueous $NaOH$ in acetone to give optically pure $(R)-(-2)-2-$ octanol, $[\alpha]_D^{25^\circ}=-10.3^\circ$. With partially racemized bromo compound whose $[\alpha]_D^{25^\circ}=+30^\circ$, the $[\alpha]_D^{25^\circ}$ of the alcohol product is -6.0°. The percentage of backside attack that took place here is $(80+x)\%$. Find the value of x.

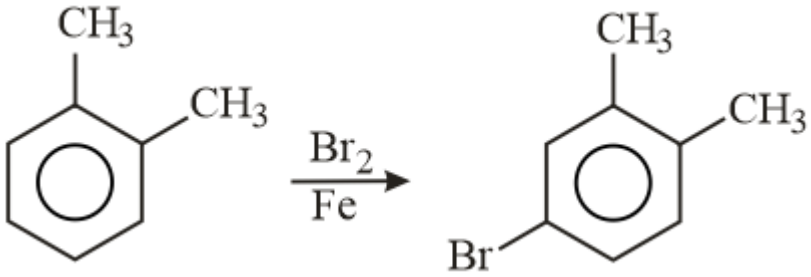
Answers & Solutions

Q.1 Answer:



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Solution:



Q.2 Answer:

The other products of the reaction are gaseous and escape out

Solution:

The reaction of $SOCl_2$ on alkanols to form alkyl chlorides gives good yields because the other products of the reaction are gaseous and escape out.

$$C_2H_5OH + Cl\text{-}SO\text{-}Cl \rightarrow C_2H_5Cl + SO_2\uparrow + HCl$$

ethanol ethylchloride

It is one of the most important methods for preparing alkyl halides.

Q.3 Answer:

C_2H_5OH, C_2H_5ONa, C_2H_5Cl

Solution:

$$2C_2H_5\text{-}OH + 2Na \rightarrow 2C_2H_5ONa + H_2$$

(A) (B)

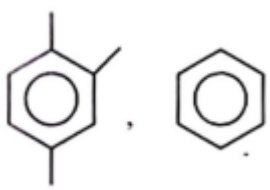
$$C_2H_5\text{-}OH + PCl_5 \rightarrow C_2H_5\text{-}Cl + POCl_3 + HCl$$

(A) (C)

$$C_2H_5\text{-}ONa + C_2H_5\text{-}Cl \rightarrow C_2H_5\text{-}O\text{-}C_2H_5 + NaCl$$

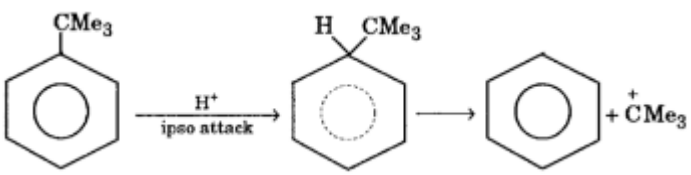
(B) (C) diethyl ether

Q.4 Answer:



Solution:

The aromatic substitution in which a substituent already present is replaced by H is called ipso substitution. i.e., reverse Friedel - Crafts reaction.



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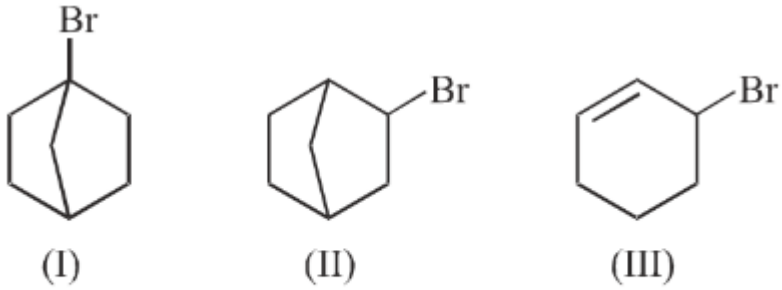
t-Butyl group is removed as stable carbocation.

Q.5 Answer:

III > II > I

Solution:

Order of SN^2 reactivity of following comp.



(I) (II) (III)

less stable corbocation due to bredt's rule

T.S. will stable due to electron cloud of p-orbital

order of SN^2 reactivity III > II > I

Q.6 Answer:

$3^\circ > 2^\circ > 1^\circ$ bromides

Solution:

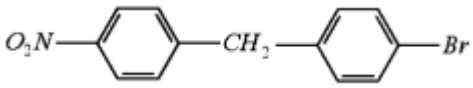
$$R - COOAg + Br_2 \xrightarrow{CCl_4} R - Br + CO_2 + AgBr \downarrow$$

The yield of alkyl bromide has the following order

Tertiary>secondary>Primary>Methyl Radical

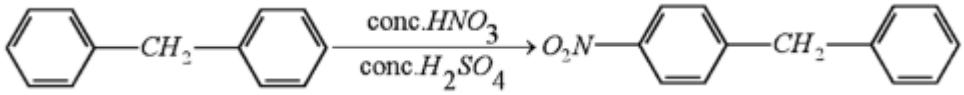
Since the mechanism of this reaction involves formation of free radicals.

Q.7 Answer:



Solution:

Originally both rings are activated and CH_2 direct the incoming electrophile to ortho/para position as:



Now, $-NO_2$ deactivates strongly to the ring on which it is present, hence bromination would occur in

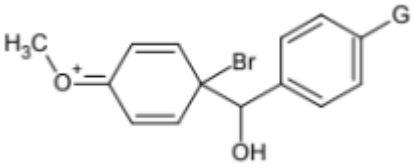
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other ring. However, $-CH_2-$ still orient the Br^+ electrophile at ortho/para position, bromination occur at less hindered para position preferably.

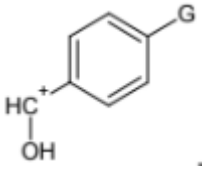
Q.8 Answer:

$-OCH_3 > -CH_3 > -H > -Br > -NO_2$

Solution:



Greater the stability of

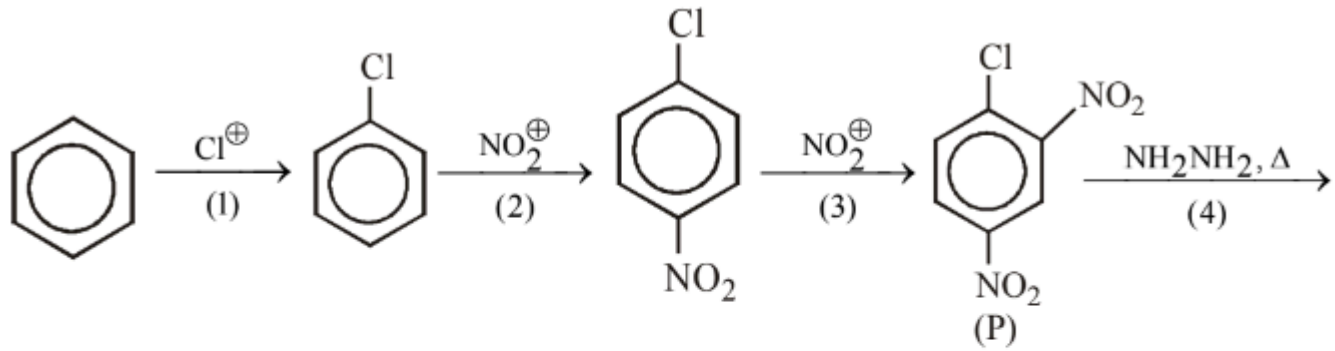


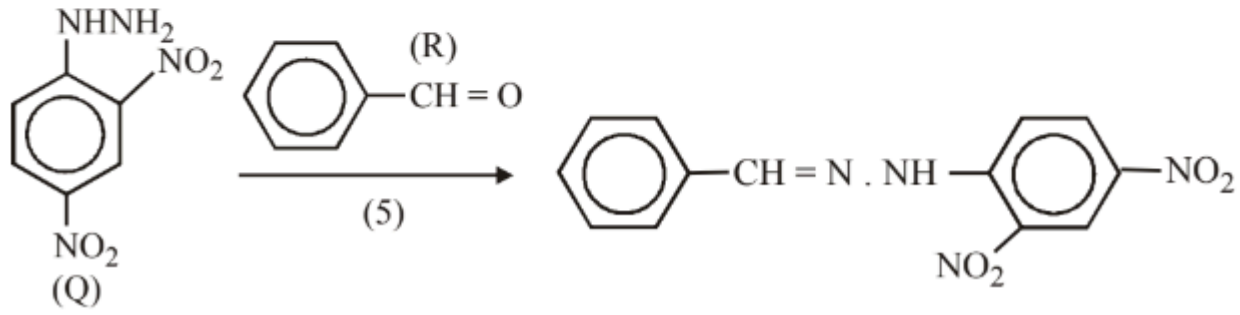
higher the leaving tendency and higher the rate of reaction.

Q.9 Answer:

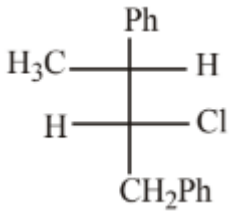
The compound Q gives a yellow precipitate in acetone

Solution:



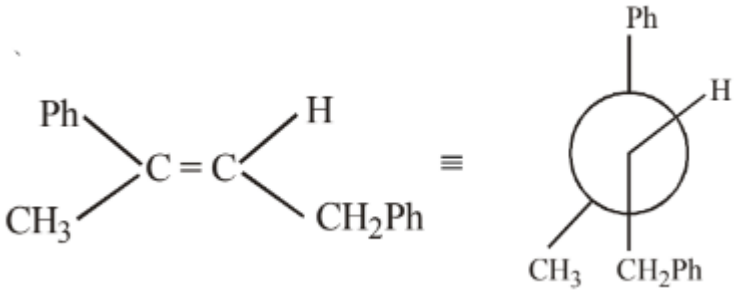


Q.10 Answer:

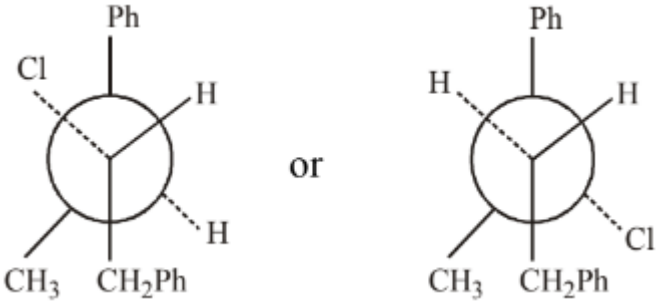


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Solution:

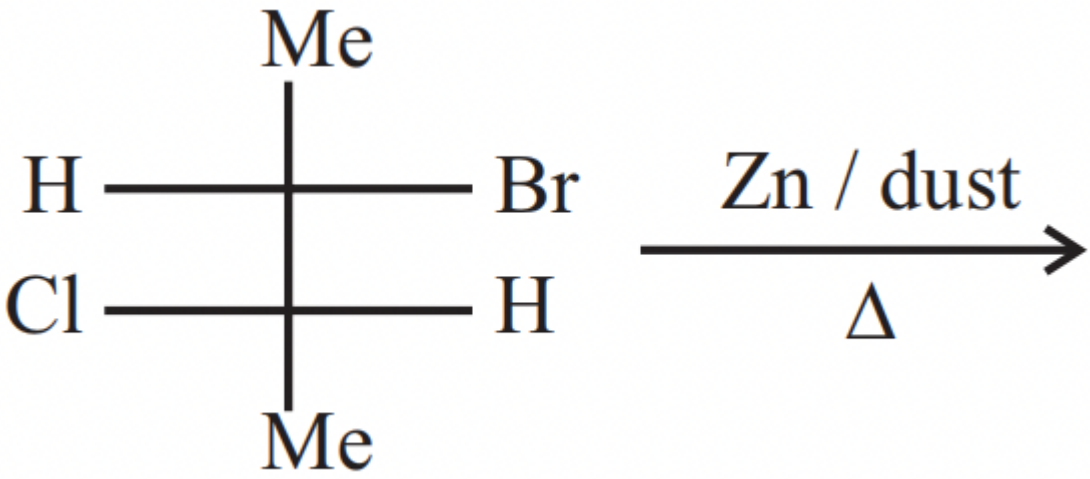


There are two possible molecules



Enantiomer give similar product.

Q.11 Answer:



Solution:

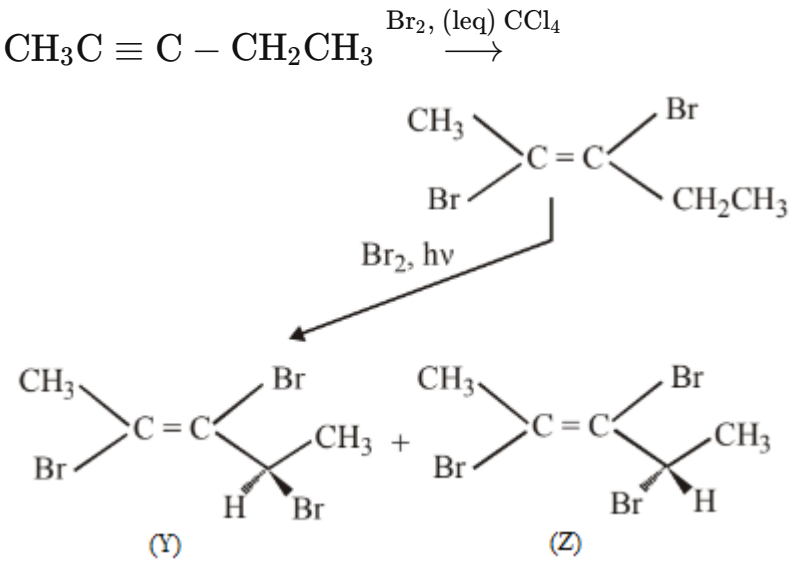
Elimination of X_2 by E_2 or zinc dust which is also antielimination and self elimination of I_2 are anti. Copes elimination is syn elimination.

Q.12 Answer:

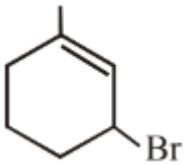
3 enantiomers

Solution:

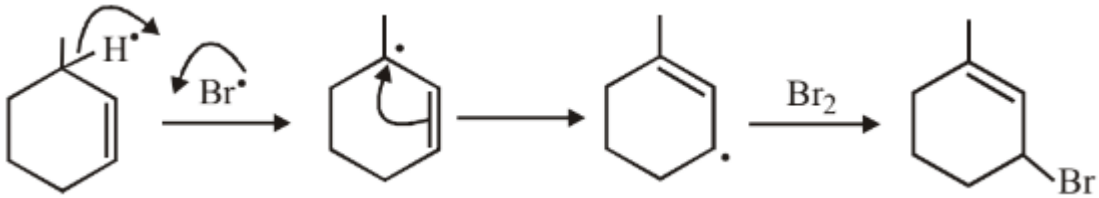
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Q.13 Answer:



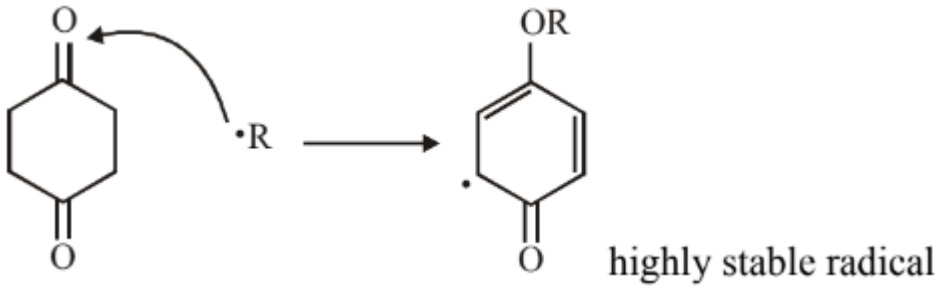
Solution:



Q.14 Answer:

benzoquinone combines with radical and produces highly stable non-productive radical

Solution:



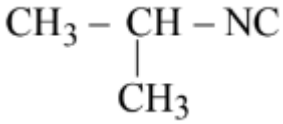
Q.15 Answer:

None of the above

Solution:

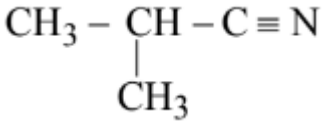
(i) 3° alkyl halide, in the presence of CN^- undergoes elimination. (CN^- is comparatively a good base).

(ii)



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S_N1 **product (more electronegative atom attack) for this solving must be** S_N1 **promoting**



S_N2 **product (less electronegative atom attack). Solvent should be** S_N2 **performing. Moreover, that should render** CN^- **comparatively less free.**

(iii) Complete S_N1 **condition.**

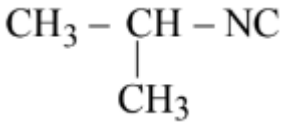
Q.16 Answer:

I. H_2O, C_2H_5OH; II. Acetone

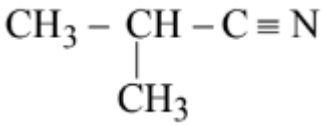
Solution:

(i) 3° **alkyl halide, in the presence of** CN^- **undergoes elimination.** $(CN^-$ **is comparatively a good base).**

(ii)



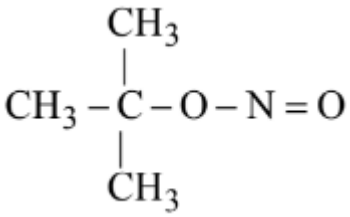
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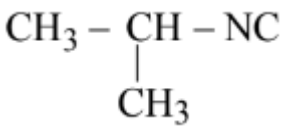
Q.17 Answer:



Solution:

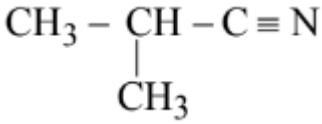
(i) 3° **alkyl halide, in the presence of** CN^- **undergoes elimination.** $(CN^-$ **is comparatively a good base).**

(ii)



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S_N1 product (more electronegative atom attack) for this solving must be S_N1 promoting



S_N2 product (less electronegative atom attack). Solvent should be S_N2 performing. Moreover, that should render CN^- comparatively less free.

(iii) Complete S_N1 condition.

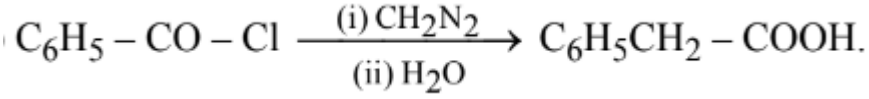
Q.18 Answer:

(A) - r,t, (B) - s, (C) - q, (D) - p

Solution:

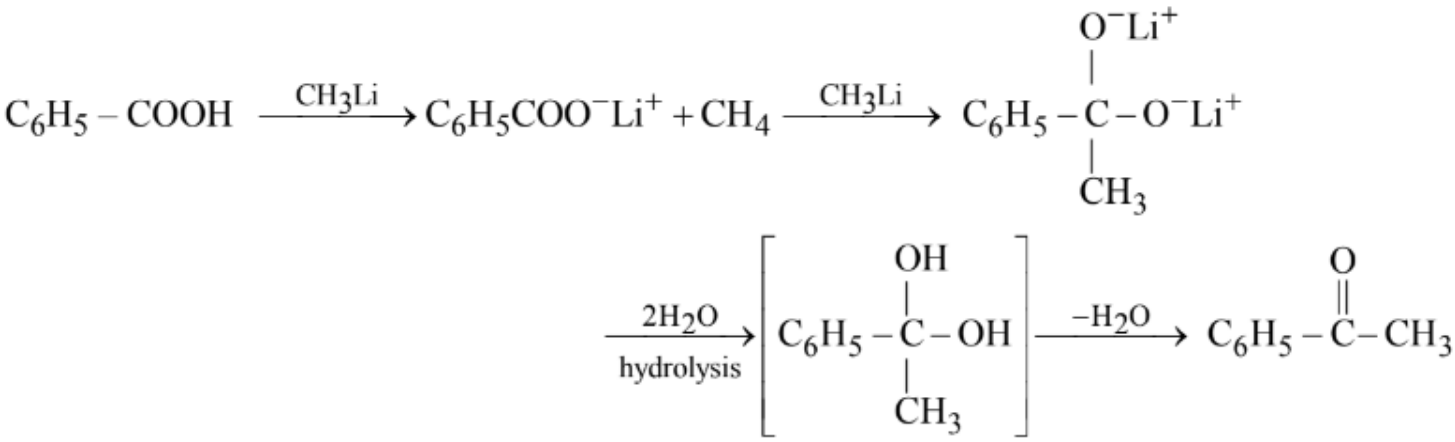
$(A) - rt, (B) - s, (C) - q, (D) - p$

(A)

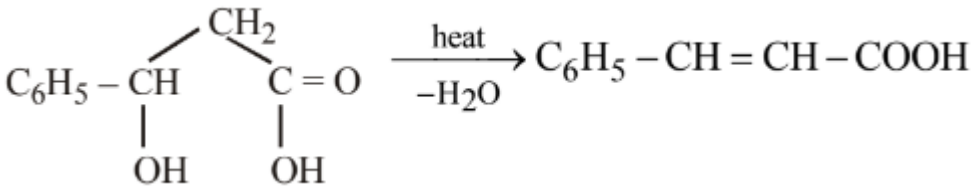


This reaction is called Arndt-Eistert reaction.

(B)

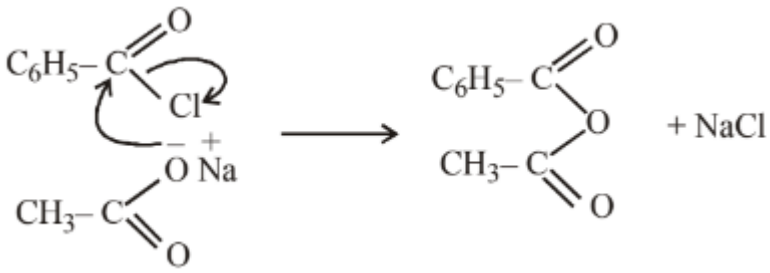


(C)



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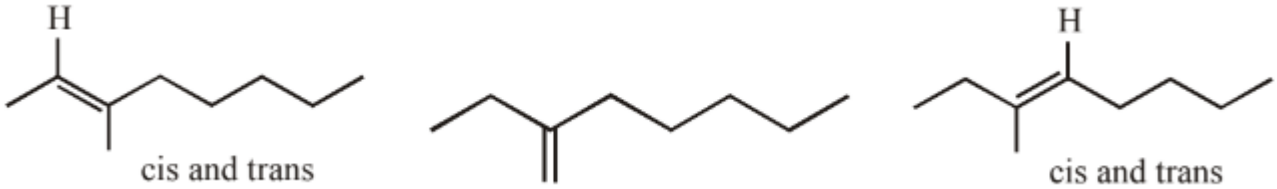
(D)



Q.19 Answer:

—

Solution:



Q.20 Answer:

—

Solution:

Optical purity of bromide $= \frac{30^\circ}{36^\circ} \times 100 = 83\%$
Optical purity of alcohol $= \frac{-6.0^\circ}{-10.3^\circ} \times 100 = 58\%$
Percentage of inversion $= \frac{58}{83} \times 100 = 70\%$
Remaining racemisation $= 30\%$
Back side attack $= 70\% + \frac{1}{2}(30\%) = 85\%$

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